This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

Device Specifications

Device Type: Audio Processor
Manufacturer: ClearOne
Firmware Version: N/A
Model(s): PSR1212, XAP 400, XAP 800, XAP TH2

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scripter Version
3.02.0000-b013	2.2.0

Version History

Module Version	Date	Notes
1_2_0_0	4/16/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  `Example: InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  `Example: InterfaceName.connectionCounter = 5`
- **0.3 second delay is REQUIRED between each update function call.**

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='PSR1212')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='PSR1212')
```

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command **AutoAnswer** [1, 2]	Value 'Off'	Value 'On'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '9' [1]	Qualifier Value 'A' – 'F' [1]	Qualifier Value '0' – '7' [2]
`# AutoAnswer example` `InterfaceName.Set('AutoAnswer', 'Off', {'Device ID': '0'})`			
Command **DTMF** [1, 2]	Value '0' – '9' 'C' '#'	Value 'A' 'D' ','	Value 'B' '*'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '9' [1]	Qualifier Value 'A' – 'F' [1]	Qualifier Value '0' – '7' [2]
`# DTMF example` `InterfaceName.Set('DTMF', '0', {'Device ID': '0'})`			
Command **Hook** [1, 2]	Value 'Off' 'Flash'	Value 'On' 'Redial'	Value 'Toggle' 'Dial'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '9' [1]	Qualifier Value 'A' – 'F' [1]	Qualifier Value '0' – '7' [2]
Qualifier Key **'Number'** [5]	Qualifier Value String		

`# Hook example - Dial value`
`InterfaceName.Set('Hook', 'Dial', {'Device ID': '0', 'Number': String})`
 	
`# Hook example - other values (not Dial)`
`InterfaceName.Set('Hook', 'On', {'Device ID': '0'})`			

Command **InputGain** [2, 3, 4]	Value -65 to 20 in steps of 0.1 dB		
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Input'**	Qualifier Value '1' – '8' [2]	Qualifier Value '1' – '12' [3, 4]	
`# InputGain example` `InterfaceName.Set('InputGain', 20, {'Device ID': '0', 'Input': '1'})`			
Command **InputMute** [2, 3, 4]	Value 'On'	Value 'Off'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Input'**	Qualifier Value '1' – '8' [2]	Qualifier Value '1' – '12' [3, 4]	
`# InputMute example` `InterfaceName.Set('InputMute', 'On', {'Device ID': '0', 'Input': '1'})`			
Command **LineInputGain** [2, 3, 4]	Value -65 to 20 in steps of 0.1 dB		
Qualifier Key	Qualifier Value		

'Device ID'	'0' – '7'		
Qualifier Key **'LineInput'**	Qualifier Value '5' – '8' [2]	Qualifier Value '9' – '12' [3, 4]	

```
# LineInputGain example
InterfaceName.Set('LineInputGain', 20, {'Device ID': '0', 'LineInput': '5'})
```

Command **LineInputMute** [2, 3, 4]	Value 'On'	Value 'Off'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'LineInput'**	Qualifier Value '5' – '8' [2]	Qualifier Value '9' – '12' [3, 4]	

```
# LineInputMute example
InterfaceName.Set('LineInputMute', 'On', {'Device ID': '0', 'LineInput': '5'})
```

Command **Macro** [2, 3, 4]	Value 1 – 255		
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		

```
# Macro example
InterfaceName.Set('Macro', 255, {'Device ID': '0'})
```

Command **Matrix** [2, 3, 4]	Value 'Off' 'Non-Gated'	Value 'On' 'Gated'	Value 'Toggle'
Qualifier Key **'Destination'**	Qualifier Value '1' – '12' [3, 4]	Qualifier Value 'A' – 'Z'	Qualifier Value '1' – '8' [2]
Qualifier Key **'DestinationGroup'**	Qualifier Value 'Outputs'	Qualifier Value 'Processing'	Qualifier Value 'ExpansionBus'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Source'**	Qualifier Value '1' – '12' [3, 4]	Qualifier Value 'A' – 'Z'	Qualifier Value '1' – '8' [2]
Qualifier Key **'SourceGroup'**	Qualifier Value 'Inputs' 'LineInputs'	Qualifier Value 'Mics' 'ExpansionBus'	Qualifier Value 'Processing'

```
# Matrix example
InterfaceName.Set('Matrix', 'Off', {'Destination': '1', 'DestinationGroup': 'Outputs', 'Device ID': '0', 'Source': '1', 'SourceGroup': 'Inputs'})
```

Command **MicGain** [2, 3, 4]	Value -65 to 20 in steps of 0.1 dB		
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Mic'**	Qualifier Value '1' – '8' [3, 4]	Qualifier Value '1' – '4' [2]	

```
# MicGain example
InterfaceName.Set('MicGain', 20, {'Device ID': '0', 'Mic': '1'})
```

Command **MicMute** [2, 3, 4]	Value 'On'	Value 'Off'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Mic'**	Qualifier Value '1' – '8' [3, 4]	Qualifier Value '1' – '4' [2]	

```
# MicMute example
InterfaceName.Set('MicMute', 'On', {'Device ID': '0', 'Mic': '1'})
```

Command	Value

OutputGain [2,3,4]	-65 to 20 in steps of 0.1 dB		
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Output'**	Qualifier Value '1' – '12' [3,4]	Qualifier Value '1' – '8' [2]	
`# OutputGain example` `InterfaceName.Set('OutputGain', 20, {'Device ID': '0', 'Output': '1'})`			
Command **OutputMute** [2,3,4]	Value 'On'	Value 'Off'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Output'**	Qualifier Value '1' – '12' [3,4]	Qualifier Value '1' – '8' [2]	
`# OutputMute example` `InterfaceName.Set('OutputMute', 'On', {'Device ID': '0', 'Output': '1'})`			
Command **Preset** [2,3,4]	Value 'Off'	Value 'Execute On'	Value 'Execute Off'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'PresetNumber'**	Qualifier Value '1' – '32'		
`# Preset example` `InterfaceName.Set('Preset', 'Off', {'Device ID': '0', 'PresetNumber': '1'})`			
Command **ProcessingChannelGain** [2,3,4]	Value -65 to 20 in steps of 0.1 dB		
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'ProcessingChannel'**	Qualifier Value 'A' – 'H' [3,4]	Qualifier Value 'A' – 'D' [2]	
`# ProcessingChannelGain example` `InterfaceName.Set('ProcessingChannelGain', 20, {'Device ID': '0', 'ProcessingChannel': 'A'})`			
Command **ProcessingChannelMute** [2,3,4]	Value 'On'	Value 'Off'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'ProcessingChannel'**	Qualifier Value 'A' – 'H' [3,4]	Qualifier Value 'A' – 'D' [2]	
`# ProcessingChannelMute example` `InterfaceName.Set('ProcessingChannelMute', 'On', {'Device ID': '0', 'ProcessingChannel': 'A'})`			
Command **Ramp** [2,3,4]	Value -65.0 to 20.0 in steps of 0.1 dB		
Qualifier Key **'Channel'**	Qualifier Value '1' – '12' [3,4] 'A' – 'D' [2]	Qualifier Value 'A' – 'H' [3,4]	Qualifier Value '1' – '9' [2]
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Group'**	Qualifier Value 'Inputs' 'LineInputs'	Qualifier Value 'Outputs' 'Processing'	Qualifier Value 'Mics'
Qualifier Key **'Rate'**	Qualifier Value -50 to 50 in steps of 1 dB/s		

`# Ramp example` `InterfaceName.Set('Ramp', 20, {'Channel': '1', 'Device ID': '0', 'Group': 'Inputs', 'Rate': 50})`			
Command **ReceiveMute** [1, 2]	Value 'On'	Value 'Off'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '9' [1]	Qualifier Value 'A' – 'F' [1]	Qualifier Value '0' – '7' [2]
`# ReceiveMute example` `InterfaceName.Set('ReceiveMute', 'On', {'Device ID': '0'})`			
Command **Ringer** [1, 2]	Value 'On'	Value 'Off'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '9' [1]	Qualifier Value 'A' – 'F' [1]	Qualifier Value '0' – '7' [2]
`# Ringer example` `InterfaceName.Set('Ringer', 'On', {'Device ID': '0'})`			
Command **SafetyMute** [2, 3, 4]	Value 'On'	Value 'Off'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
`# SafetyMute example` `InterfaceName.Set('SafetyMute', 'On', {'Device ID': '0'})`			
Command **SpeedDial** [1, 2]	Value '1' – '10'		
Qualifier Key **'Device ID'**	Qualifier Value '0' – '9' [1]	Qualifier Value 'A' – 'F' [1]	Qualifier Value '0' – '7' [2]
`# SpeedDial example` `InterfaceName.Set('SpeedDial', '1', {'Device ID': '0'})`			
Command **StringExecution**	Value '0' – '7'		
Qualifier Key **'Device ID'**	Qualifier Value '0' – '9' [1]	Qualifier Value 'A' – 'F' [1]	Qualifier Value '0' – '7' [2, 3, 4]
`# StringExecution example` `InterfaceName.Set('StringExecution', '0', {'Device ID': '0'})`			
Command **TransmitMute** [1, 2]	Value 'On'	Value 'Off'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '9' [1]	Qualifier Value 'A' – 'F' [1]	Qualifier Value '0' – '7' [2]
`# TransmitMute example` `InterfaceName.Set('TransmitMute', 'On', {'Device ID': '0'})`			

[1] only available for XAP TH2
[2] only available for XAP 400
[3] only available for XAP 800
[4] only available for PSR1212
[5] only needed for 'Dial' command.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
        InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
        Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
        InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
FeedbackHandler)
        FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
        InterfaceName.Update(Command)
        Value = InterfaceName.ReadStatus(Command)
        InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
        FeedbackHandler will be called when any qualifier gets a new status.
```

Command **AutoAnswer** [1, 2]	Value 'Off'	Value 'On'	Value 'Toggle'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '9' [1]	Qualifier Value 'A' – 'F' [1]	Qualifier Value '0' – '7' [2]

`# AutoAnswer example`
`InterfaceName.Update('AutoAnswer', {'Device ID': '0'})`
`Value = InterfaceName.ReadStatus('AutoAnswer', {'Device ID': '0'})`
`InterfaceName.SubscribeStatus('AutoAnswer', None, FeedbackHandler)`			

Command **ConnectionStatus**	Value 'Connected'	Value 'Disconnected'	
`Value = InterfaceName.ReadStatus('ConnectionStatus')` `InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)`			
Command **Hook** [1, 2]	Value 'Off' 'Flash' 'Dial'	Value 'On' 'Redial'	Value 'Toggle' 'Ringing'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '9' [1]	Qualifier Value 'A' – 'F' [1]	Qualifier Value '0' – '7' [2]

`# Hook example`
`InterfaceName.Update('Hook', {'Device ID': '0'})`
`Value = InterfaceName.ReadStatus('Hook', {'Device ID': '0'})`
`InterfaceName.SubscribeStatus('Hook', None, FeedbackHandler)`			

Command **InputGain** [2, 3, 4]	Value -65 to 20 in steps of 0.1 dB		
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Input'**	Qualifier Value '1' – '8' [2]	Qualifier Value '1' – '12' [3, 4]	

`# InputGain example`
`InterfaceName.Update('InputGain', {'Device ID': '0', 'Input': '1'})`
`Value = InterfaceName.ReadStatus('InputGain', {'Device ID': '0', 'Input': '1'})`
`InterfaceName.SubscribeStatus('InputGain', None, FeedbackHandler)`			

Command **InputMute** [2, 3, 4]	Value 'On'	Value 'Off'	
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Input'**	Qualifier Value '1' – '8' [2]	Qualifier Value '1' – '12' [3, 4]	

```
    # InputMute example
    InterfaceName.Update('InputMute', {'Device ID': '0', 'Input': '1'})
    Value = InterfaceName.ReadStatus('InputMute', {'Device ID': '0', 'Input': '1'})
    InterfaceName.SubscribeStatus('InputMute', None, FeedbackHandler)
```

Command **LineInputGain** [2, 3, 4]	Value -65 to 20 in steps of 0.1 dB		
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'LineInput'**	Qualifier Value '5' – '8' [2]	Qualifier Value '9' – '12' [3,4]	

```
    # LineInputGain example
    InterfaceName.Update('LineInputGain', {'Device ID': '0', 'LineInput': '5'})
    Value = InterfaceName.ReadStatus('LineInputGain', {'Device ID': '0', 'LineInput': '5'})
    InterfaceName.SubscribeStatus('LineInputGain', None, FeedbackHandler)
```

Command **LineInputMute** [2, 3, 4]	Value 'On'	Value 'Off'	
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'LineInput'**	Qualifier Value '5' – '8' [2]	Qualifier Value '9' – '12' [3,4]	

```
    # LineInputMute example
    InterfaceName.Update('LineInputMute', {'Device ID': '0', 'LineInput': '5'})
    Value = InterfaceName.ReadStatus('LineInputMute', {'Device ID': '0', 'LineInput': '5'})
    InterfaceName.SubscribeStatus('LineInputMute', None, FeedbackHandler)
```

Command **Macro** [2, 3, 4]	Value 1 – 255		
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		

```
    # Macro example
    InterfaceName.Update('Macro', {'Device ID': '0'})
    Value = InterfaceName.ReadStatus('Macro', {'Device ID': '0'})
    InterfaceName.SubscribeStatus('Macro', None, FeedbackHandler)
```

Command **Matrix** [2, 3, 4]	Value 'Off' 'Non-Gated'	Value 'On' 'Gated'	Value 'Toggle'
Qualifier Key **'Destination'**	Qualifier Value '1' – '12' [3,4]	Qualifier Value 'A' – 'Z'	Qualifier Value '1' – '8' [2]
Qualifier Key **'DestinationGroup'**	Qualifier Value 'Outputs'	Qualifier Value 'Processing'	Qualifier Value 'ExpansionBus'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'Source'**	Qualifier Value '1' – '12' [3,4]	Qualifier Value 'A' – 'Z'	Qualifier Value '1' – '8' [2]
Qualifier Key **'SourceGroup'**	Qualifier Value 'Inputs' 'LineInputs'	Qualifier Value 'Mics' 'ExpansionBus'	Qualifier Value 'Processing'

```
    # Matrix example
    InterfaceName.Update('Matrix', {'Destination': '1', 'DestinationGroup': 'Outputs', 'Device 
ID': '0', 'Source': '1', 'SourceGroup': 'Inputs'})
    Value = InterfaceName.ReadStatus('Matrix', {'Destination': '1', 'DestinationGroup': 
'Outputs', 'Device ID': '0', 'Source': '1', 'SourceGroup': 'Inputs'})
    InterfaceName.SubscribeStatus('Matrix', None, FeedbackHandler)
```

Command **MicGain** [2, 3, 4]	Value -65 to 20 in steps of 0.1 dB
Qualifier Key	Qualifier Value

'Device ID'	'0' – '7'	
Qualifier Key **'Mic'**	Qualifier Value '1' – '8' [3,4]	Qualifier Value '1' – '4' [2]

```
# MicGain example
InterfaceName.Update('MicGain', {'Device ID': '0', 'Mic': '1'})
Value = InterfaceName.ReadStatus('MicGain', {'Device ID': '0', 'Mic': '1'})
InterfaceName.SubscribeStatus('MicGain', None, FeedbackHandler)
```

Command **MicMute** [2,3,4]	Value 'On'	Value 'Off'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'	
Qualifier Key **'Mic'**	Qualifier Value '1' – '8' [3,4]	Qualifier Value '1' – '4' [2]

```
# MicMute example
InterfaceName.Update('MicMute', {'Device ID': '0', 'Mic': '1'})
Value = InterfaceName.ReadStatus('MicMute', {'Device ID': '0', 'Mic': '1'})
InterfaceName.SubscribeStatus('MicMute', None, FeedbackHandler)
```

Command **OutputGain** [2,3,4]	Value -65 to 20 in steps of 0.1 dB	
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'	
Qualifier Key **'Output'**	Qualifier Value '1' – '12' [3,4]	Qualifier Value '1' – '8' [2]

```
# OutputGain example
InterfaceName.Update('OutputGain', {'Device ID': '0', 'Output': '1'})
Value = InterfaceName.ReadStatus('OutputGain', {'Device ID': '0', 'Output': '1'})
InterfaceName.SubscribeStatus('OutputGain', None, FeedbackHandler)
```

Command **OutputMute** [2,3,4]	Value 'On'	Value 'Off'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'	
Qualifier Key **'Output'**	Qualifier Value '1' – '12' [3,4]	Qualifier Value '1' – '8' [2]

```
# OutputMute example
InterfaceName.Update('OutputMute', {'Device ID': '0', 'Output': '1'})
Value = InterfaceName.ReadStatus('OutputMute', {'Device ID': '0', 'Output': '1'})
InterfaceName.SubscribeStatus('OutputMute', None, FeedbackHandler)
```

Command **Preset** [2,3,4]	Value 'Off'	Value 'Execute On'	Value 'Execute Off'
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'		
Qualifier Key **'PresetNumber'**	Qualifier Value '1' – '32'		

```
# Preset example
InterfaceName.Update('Preset', {'Device ID': '0', 'PresetNumber': '1'})
Value = InterfaceName.ReadStatus('Preset', {'Device ID': '0', 'PresetNumber': '1'})
InterfaceName.SubscribeStatus('Preset', None, FeedbackHandler)
```

Command **ProcessingChannelGain** [2,3,4]	Value -65 to 20 in steps of 0.1 dB	
Qualifier Key **'Device ID'**	Qualifier Value '0' – '7'	
Qualifier Key **'ProcessingChannel'**	Qualifier Value 'A' – 'H' [3,4]	Qualifier Value 'A' – 'D' [2]

```
    # ProcessingChannelGain example
    InterfaceName.Update('ProcessingChannelGain', {'Device ID': '0', 'ProcessingChannel': 'A'})
    Value = InterfaceName.ReadStatus('ProcessingChannelGain', {'Device ID': '0',
'ProcessingChannel': 'A'})
    InterfaceName.SubscribeStatus('ProcessingChannelGain', None, FeedbackHandler)
```

Command	Value	Value	Value
ProcessingChannelMute [2, 3, 4]	'On'	'Off'	'Toggle'
Qualifier Key	Qualifier Value		
'Device ID'	'0' – '7'		
Qualifier Key	Qualifier Value	Qualifier Value	
'ProcessingChannel'	'A' – 'H' [3, 4]	'A' – 'D' [2]	

```
    # ProcessingChannelMute example
    InterfaceName.Update('ProcessingChannelMute', {'Device ID': '0', 'ProcessingChannel': 'A'})
    Value = InterfaceName.ReadStatus('ProcessingChannelMute', {'Device ID': '0',
'ProcessingChannel': 'A'})
    InterfaceName.SubscribeStatus('ProcessingChannelMute', None, FeedbackHandler)
```

Command	Value	Value	
ReceiveMute [1, 2]	'On'	'Off'	
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Device ID'	'0' – '9' [1]	'A' – 'F' [1]	'0' – '7' [2]

```
    # ReceiveMute example
    InterfaceName.Update('ReceiveMute', {'Device ID': '0'})
    Value = InterfaceName.ReadStatus('ReceiveMute', {'Device ID': '0'})
    InterfaceName.SubscribeStatus('ReceiveMute', None, FeedbackHandler)
```

Command	Value	Value	
Ringer [1, 2]	'On'	'Off'	
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Device ID'	'0' – '9' [1]	'A' – 'F' [1]	'0' – '7' [2]

```
    # Ringer example
    InterfaceName.Update('Ringer', {'Device ID': '0'})
    Value = InterfaceName.ReadStatus('Ringer', {'Device ID': '0'})
    InterfaceName.SubscribeStatus('Ringer', None, FeedbackHandler)
```

Command	Value	Value
SafetyMute [2, 3, 4]	'On'	'Off'
Qualifier Key	Qualifier Value	
'Device ID'	'0' – '7'	

```
    # SafetyMute example
    InterfaceName.Update('SafetyMute', {'Device ID': '0'})
    Value = InterfaceName.ReadStatus('SafetyMute', {'Device ID': '0'})
    InterfaceName.SubscribeStatus('SafetyMute', None, FeedbackHandler)
```

Command	Value		
SpeedDial [1, 2]	'1' – '10'		
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Device ID'	'0' – '9' [1]	'A' – 'F' [1]	'0' – '7' [2]

```
    # SpeedDial example
    InterfaceName.Update('SpeedDial', {'Device ID': '0'})
    Value = InterfaceName.ReadStatus('SpeedDial', {'Device ID': '0'})
    InterfaceName.SubscribeStatus('SpeedDial', None, FeedbackHandler)
```

Command	Value		
StringExecution	'0' – '7'		
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Device ID'	'0' – '9' [1]	'A' – 'F' [1]	'0' – '7' [2, 3 4]

```
    # StringExecution example
    InterfaceName.Update('StringExecution', {'Device ID': '0'})
    Value = InterfaceName.ReadStatus('StringExecution', {'Device ID': '0'})
```

```
InterfaceName.SubscribeStatus('StringExecution', None, FeedbackHandler)
```

Command	Value	Value	Value
TransmitMute [1, 2]	'On'	'Off'	'Toggle'
Qualifier Key	**Qualifier Value**	**Qualifier Value**	**Qualifier Value**
'Device ID'	'0' – '9' [1]	'A' – 'F' [1]	'0' – '7' [2]

```
# TransmitMute example
InterfaceName.Update('TransmitMute', {'Device ID': '0'})
Value = InterfaceName.ReadStatus('TransmitMute', {'Device ID': '0'})
InterfaceName.SubscribeStatus('TransmitMute', None, FeedbackHandler)
```

Command	Value
Version	'String'

```
# Version example
InterfaceName.Update('Version')
Value = InterfaceName.ReadStatus('Version')
InterfaceName.SubscribeStatus('Version', None, FeedbackHandler)
```

[1] only available for XAP TH2
[2] only available for XAP 400
[3] only available for XAP 800
[4] only available for PSR1212

Cable and Adapter Requirements

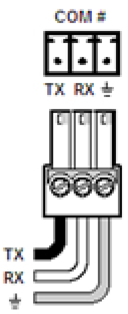
Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

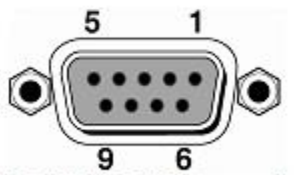
Serial communication

Port Type:	RS-232	**Parity:**	None
Baud Rate:	38400	**Stop Bits:**	One
Data Bits:	8	**Flow Control:**	None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



RS232 DB9, Female
Audio Processor

Appendix A. Set Commands

Each command string requires a device type character and a device ID character.

Device Type is denoted by {0} in the example strings below. Replace {0} with the string character listed below per model.

- PSR1212 = '4'
- XAP 400 = '7'
- XAP 800 = '5'
- XAP TH2 = '6'

Device ID is denoted by {1} in the example strings below. Replace {1} with the string character listed below per model.

- PSR1212 = '0' – '7'
- XAP 400 = '0' – '7'
- XAP 800 = '0' – '7'
- XAP TH2 = '0' – '9' or 'A' – 'F'

Auto Answer Off	`#{0}{1} AA 1 0\x0D`
Auto Answer On	`#{0}{1} AA 1 1\x0D`
Auto Answer Toggle	`#{0}{1} AA 1 2\x0D`
DTMF 0	`#{0}{1} DIAL 1 0\x0D`
DTMF 1	`#{0}{1} DIAL 1 1\x0D`
DTMF 2	`#{0}{1} DIAL 1 2\x0D`
DTMF 3	`#{0}{1} DIAL 1 3\x0D`
DTMF 4	`#{0}{1} DIAL 1 4\x0D`
DTMF 5	`#{0}{1} DIAL 1 5\x0D`
DTMF 6	`#{0}{1} DIAL 1 6\x0D`
DTMF 7	`#{0}{1} DIAL 1 7\x0D`
DTMF 8	`#{0}{1} DIAL 1 8\x0D`
DTMF 9	`#{0}{1} DIAL 1 9\x0D`
DTMF A	`#{0}{1} DIAL 1 A\x0D`
DTMF B	`#{0}{1} DIAL 1 B\x0D`
DTMF C	`#{0}{1} DIAL 1 C\x0D`
DTMF D	`#{0}{1} DIAL 1 D\x0D`
DTMF *	`#{0}{1} DIAL 1 *\x0D`
DTMF #	`#{0}{1} DIAL 1 #\x0D`
DTMF ,	`#{0}{1} DIAL 1 ,\x0D`
Hook On	`#{0}{1} TE 1 0\x0D`
Hook Off	`#{0}{1} TE 1 1\x0D`
Hook Toggle	`#{0}{1} TE 1 2\x0D`
Hook Flash	`#{0}{1} HOOK 1\x0D`
Hook Redial	`#{0}{1} REDIAL 1\x0D`
Hook Dial Number String	`#{0}{1} DIAL 1 String\x0D`
Input Gain Input 1 -65.0	`#{0}{1} GAIN 1 I -65.0 A\x0D`
Input Gain Input 8 20.0	`#{0}{1} GAIN 8 I 20.0 A\x0D`

Input Mute Input 1 On	#{0}{1} MUTE 1 I 1\x0D
Input Mute Input 1 Off	#{0}{1} MUTE 1 I 0\x0D
Input Mute Input 1 Toggle	#{0}{1} MUTE 1 I 2\x0D
Input Mute Input 12 On	#{0}{1} MUTE 12 I 1\x0D
Input Mute Input 12 Off	#{0}{1} MUTE 12 I 0\x0D
Input Mute Input 12 Toggle	#{0}{1} MUTE 12 I 2\x0D
Line Input Gain LineInput 5 -65.0	#{0}{1} GAIN 5 L -65.0 A\x0D
Line Input Gain LineInput 9 20.0	#{0}{1} GAIN 9 L 20.0 A\x0D
Line Input Mute LineInput 5 On	#{0}{1} MUTE 5 L 1\x0D
Line Input Mute LineInput 5 Off	#{0}{1} MUTE 5 L 0\x0D
Line Input Mute LineInput 5 Toggle	#{0}{1} MUTE 5 L 2\x0D
Line Input Mute LineInput 9 On	#{0}{1} MUTE 9 L 1\x0D
Line Input Mute LineInput 9 On	#{0}{1} MUTE 9 L 0\x0D
Line Input Mute LineInput 9 On	#{0}{1} MUTE 5 L 2\x0D
Macro 1	#{0}{1} MACRO 1\x0D
Macro 255	#{0}{1} MACRO 255\x0D
Matrix (see Appendix B)	#{0}{1} MTRX {2} {3} {4} {5} {6}\x0D
Mic Gain Mic 1 -65.0	#{0}{1} GAIN 1 M -65.0 A\x0D
Mic Gain Mic 8 20.0	#{0}{1} GAIN 8 M 20.0 A\x0D
Mic Mute Mic 1 On	#{0}{1} MUTE 1 M 1\x0D
Mic Mute Mic 1 Off	#{0}{1} MUTE 1 M 0\x0D
Mic Mute Mic 1 Toggle	#{0}{1} MUTE 1 M 2\x0D
Mic Mute Mic 8 On	#{0}{1} MUTE 8 M 1\x0D
Mic Mute Mic 8 Off	#{0}{1} MUTE 8 M 0\x0D
Mic Mute Mic 8 Toggle	#{0}{1} MUTE 8 M 2\x0D
Output Gain Output 1 -65.0	#{0}{1} GAIN 1 O -65.0 A\x0D
Output Gain Output 12 20.0	#{0}{1} GAIN 12 O 20.0 A\x0D
Output Mute Output 1 On	#{0}{1} MUTE 1 O 1\x0D
Output Mute Output 1 Off	#{0}{1} MUTE 1 O 0\x0D
Output Mute Output 1 Toggle	#{0}{1} MUTE 1 O 2\x0D
Output Mute Output 12 On	#{0}{1} MUTE 12 O 1\x0D
Output Mute Output 12 Off	#{0}{1} MUTE 12 O 0\x0D
Output Mute Output 12 Toggle	#{0}{1} MUTE 12 O 2\x0D
Preset PresetNumber 1 Off	#{0}{1} PRESET 1 0\x0D
Preset PresetNumber 1 Execute On	#{0}{1} PRESET 1 1\x0D
Preset PresetNumber 1 Execute Off	#{0}{1} PRESET 1 2\x0D
Preset PresetNumber 32 Off	#{0}{1} PRESET 32 0\x0D
Preset PresetNumber 32 Execute On	#{0}{1} PRESET 32 1\x0D
Preset PresetNumber 32 Execute Off	#{0}{1} PRESET 32 2\x0D
Processing Channel Gain ProcessingChannel A -65.0	#{0}{1} GAIN A P -65.0 A\x0D
Processing Channel Gain ProcessingChannel H 20.0	#{0}{1} GAIN H P 20.0 A\x0D
Processing Channel Mute ProcessingChannel A On	#{0}{1} MUTE A P 1\x0D
Processing Channel Mute ProcessingChannel A Off	#{0}{1} MUTE A P 0\x0D
Processing Channel Mute ProcessingChannel A Toggle	#{0}{1} MUTE A P 2\x0D

Processing Channel Mute ProcessingChannel H On	#{0}{1} MUTE H P 1\x0D
Processing Channel Mute ProcessingChannel H Off	#{0}{1} MUTE A P 0\x0D
Processing Channel Mute ProcessingChannel H Toggle	#{0}{1} MUTE A P 2\x0D
Ramp (see Appendix C)	#{0}{1} RAMP {2} {3} {4} {5}\x0D
Receive Mute On	#{0}{1} MUTE 1 R 1\r
Receive Mute Off	#{0}{1} MUTE 1 R 0\r
Receive Mute Toggle	#{0}{1} MUTE 1 R 2\r
Ringer On	#{0}{1} RINGER 1 1\r
Ringer Off	#{0}{1} RINGER 1 0\r
Ringer Toggle	#{0}{1} RINGER 1 2\r
Safety Mute On	#{0}{1} SFTYMUTE 1\r
Safety Mute Off	#{0}{1} SFTYMUTE 0\r
Safety Mute Toggle	#{0}{1} SFTYMUTE 2\r
Speed Dial 1	#{0}{1} SPEEDDIAL 1 1\r
Speed Dial 10	#{0}{1} SPEEDDIAL 1 10\r
String Execution 0	#{0}{1} STRING 0\r
String Execution 7	#{0}{1} STRING 7\r
Transmit Mute On	#{0}{1} MUTE 1 T 1\r
Transmit Mute Off	#{0}{1} MUTE 1 T 0\r
Transmit Mute Toggle	#{0}{1} MUTE 1 T 2\r

Appendix B. Matrix Command

Matrix set command
`#{0}{1} MTRX {2} {3} {4} {5} {6}\x0D`

Matrix update command
`#{0}{1} MTRX {2} {3} {4} {5}\x0D`

{0} – Device Type
{1} – Device ID
{2} – Source: '1' to '12' or 'A' to 'Z'
{3} – SourceGroup: 'I' for Inputs, 'M' for Mics, 'P' for Processing, 'L' for LineInputs, 'E' for ExpansionBus
{4} – Destination: '1' to '12' or 'A' to 'Z'
{5} – DestinationGroup: 'O' for Outputs, 'P' for Processing, 'E' for ExpansionBus
{6} – Value: '0' for Off, '1' for On, '2' for Toggle, '3' for Non-Gated, '4' for Gated

Example

Matrix set command for PSR1212, Device ID '0', Source '1', SourceGroup 'Inputs', Destination '1', DestinationGroup 'Outputs', and value 'On' would be:

```
#40 MTRX 1 I 1 O 2\x0D
```

Appendix C. Ramp Command

Ramp set command
`#{0}{1} RAMP {2} {3} {4} {5}\x0D`

{0} – Device Type
{1} – Device ID
{2} – Channel: '1' to '12' or 'A' to 'H'
{3} – Group: 'I' for Inputs, 'O' for Outputs, 'M' for Mics, 'L' for LineInputs, 'P' for Processing
{4} – Rate: -50 to 50
{5} – Value: -65.0 to 20.0 in steps of 0.1

Example

Ramp for PSR1212, Device ID '0', Channel '1', Group 'Inputs', Rate -10, and value -35.5.

```
#40 RAMP 1 I -10 -35.5\x0D
```

Appendix D. Update Commands

Each command string requires a device type character and a device ID character.

Device Type is denoted by {0} in the example strings below. Replace {0} with the string character listed below per model.

- PSR1212 = '4'
- XAP 400 = '7'
- XAP 800 = '5'
- XAP TH2 = '6'

Device ID is denoted by {1} in the example strings below. Replace {1} with the string character listed below per model.

- PSR1212 = '0' – '7'
- XAP 400 = '0' – '7'
- XAP 800 = '0' – '7'
- XAP TH2 = '0' – '9' or 'A' – 'F'

Auto Answer	`#{0}{1} AA 1\r`
Hook	`#{0}{1} TE 1\r`
Input Gain Input 1	`#{0}{1} GAIN 1 I\r`
Input Gain Input 12	`#{0}{1} GAIN 12 I\r`
Input Mute Input 1	`#{0}{1} MUTE 1 I\r`
Input Mute Input 12	`#{0}{1} MUTE 12 I\r`
Line Input Gain LineInput 5	`#{0}{1} GAIN 5 L\r`
Line Input Gain LineInput 9	`#{0}{1} GAIN 9 L\r`
Line Input Mute LineInput 5	`#{0}{1} MUTE 5 L\r`
Line Input Mute LineInput 9	`#{0}{1} MUTE 9 L\r`
Macro	`#{0}{1} MACRO\r`
Matrix (see appendix B)	`#{0}{1} MTRX {2} {3} {4} {5}\r`
Mic Gain Mic 1	`#{0}{1} GAIN 1 M\r`
Mic Gain Mic 8	`#{0}{1} GAIN 8 M\r`
Mic Mute Mic 1	`#{0}{1} MUTE 1 M\r`
Mic Mute Mic 8	`#{0}{1} MUTE 8 M\r`
Output Gain Output 1	`#{0}{1} GAIN 1 O\r`
Output Gain Output 12	`#{0}{1} GAIN 12 O\r`
Output Mute Output 1	`#{0}{1} MUTE 1 O\r`
Output Mute Output 12	`#{0}{1} MUTE 12 O\r`
Preset PresetNumber 1	`#{0}{1} PRESET 1\r`
Preset PresetNumber 32	`#{0}{1} PRESET 32\r`
Processing Channel Gain ProcessingChannel A	`#{0}{1} GAIN A P\r`
Processing Channel Gain ProcessingChannel H	`#{0}{1} GAIN H P\r`
Processing Channel Mute ProcessingChannel A	`#{0}{1} MUTE A P\r`
Processing Channel Mute ProcessingChannel H	`#{0}{1} MUTE H P\r`

Receive Mute	#{0}{1} MUTE 1 R\r
Ringer	#{0}{1} RINGER 1\r
Safety Mute	#{0}{1} SFTYMUTE\r
Speed Dial	#{0}{1} SPEEDDIAL 1\r
String Execution	#{0}{1} STRING\r
Transmit Mute	#{0}{1} MUTE 1 T\r
Version	#{0}{1} VER\r